CMPT 3830: Project Proposal Template

1. Project Title:

Loyalty-Driven Dealer Performance Analysis: Understanding Customer Retention and Revenue Patterns Across GO Auto's Edmonton Dealerships

2. Project Overview:

- Objective:

Analyze how loyalty program participation impacts dealership performance and customer behavior across GO Auto's seven Edmonton dealerships. This project will develop machine learning clustering techniques to segment customers and dealers based on loyalty-driven behaviors, identifying whether dealerships with higher loyalty card usage demonstrate stronger customer retention, higher average revenue, and more consistent service activity. The analysis will deliver actionable insights to help GO Auto leadership optimize their loyalty program and address the critical challenge of losing customers around the five-year mark after vehicle purchase.

- Background:

GO Auto Services operates seven dealerships across Edmonton, serving a diverse customer base with varying vehicle brands, price points, and service needs. The company has implemented a loyalty card program to encourage repeat business and build long-term customer relationships.

The available dataset contains 242,818 service records spanning 2020-2024, including customer payment information, vehicle details, service dates, loyalty card status, appointment scheduling, mileage, dealer information, and customer distance traveled.

This project addresses Problem Statement #5 from the CMPT 3830 course requirements, utilizing supervised and unsupervised machine learning techniques to provide data-driven recommendations for GO Auto's business strategy.

- Scope:

In Scope:

- Clustering analysis to segment dealerships based on loyalty program performance metrics
- Customer segmentation into distinct behavior groups (loyalty-driven repeat clients, occasional users, non-loyalty walk-ins)

- Comparative analysis of dealership performance across loyalty adoption rates, revenue patterns, and service frequency
- Distance analysis examining loyalty members' willingness to travel to specific dealerships
- Feature engineering to create relevant metrics for machine learning models
- Development of interactive visualizations and a performance dashboard
- Business-oriented recommendations for improving loyalty program effectiveness and dealer performance

Out of Scope:

- Individual customer identity analysis (all analysis will be conducted at aggregate levels)
- Marketing campaign development or implementation
- Financial forecasting beyond descriptive statistics from existing data
- Integration with GO Auto's operational systems
- Analysis of non-Edmonton dealership locations [Certainty: 90% - scope boundaries align with course project requirements and dataset availability]

3. Project Deliverables:

List the specific outputs or deliverables expected from the project.

1. **Demo 1 (February 10, 2026):** Interactive presentation demonstrating initial data exploration, preliminary clustering results, and key insights from exploratory data analysis. Includes data visualization prototypes and initial customer/dealer segmentation framework.

2. **Phase 1 Report (February 27, 2026):** Comprehensive written report documenting problem understanding, data preprocessing methodology, exploratory data analysis findings, initial clustering implementation, and preliminary business insights. Includes statistical analysis of loyalty program impact on dealership performance.

3. **Demo 2 (March 10, 2026):** Presentation showcasing refined clustering models, comparative dealership performance analysis, and interactive dashboard prototype. Demonstrates customer segmentation results and distance-based insights.

4. **Phase 2 Report (March 20, 2026):** Advanced analysis report including optimized machine learning models, detailed customer behavior segmentation, dealership benchmarking results, and comprehensive recommendations for loyalty program enhancement. Includes model validation metrics and statistical significance testing.

5. **Final Demo (April 7, 2026):** Professional presentation to course instructors and stakeholders showcasing complete project outcomes, interactive performance dashboard, actionable business recommendations, and demonstration of machine learning model capabilities. Includes reflection on project learnings and potential future enhancements.

6. **GitHub Repository:** Well-documented codebase including all data preprocessing scripts, machine learning implementations, visualization code, and analysis notebooks. Maintained throughout project lifecycle with clear commit history and README documentation.

4. Project Timeline:

Break down the project into phases and include estimated completion dates.

| Milestone | Completion Date |

Milestone	Completion Date
Phase 1: Data Foundation & Initial Analysis	
• Data exploration and quality assessment	February 3, 2026
• Feature engineering and data preprocessing	February 6, 2026
• Initial clustering implementation	February 9, 2026
Demo 1 Presentation	**February 10, 2026**
• Statistical analysis and hypothesis testing	February 20, 2026
• Visualization development	February 24, 2026
Phase 1 Report Submission	**February 27, 2026**
Phase 2: Advanced Modeling & Insights	

• Cluster refinement and optimization	March 4, 2026
• Customer segmentation analysis	March 7, 2026
Demo 2 Presentation	**March 10, 2026**
• Dashboard development	March 28, 2026
• Business recommendations development	March 17, 2026
Phase 2 Report Submission	**March 20, 2026**
Phase 3: Final Integration & Presentation	
• Final presentation preparation	April 5, 2026
Final Demo Presentation	**April 7, 2026**

Note: Weekly Sunday sync meetings (7:00-8:30 PM MST) occur throughout all phases for progress review and planning.

5. Project Plan:

- Tasks and Activities:

Provide a breakdown of the tasks needed to complete the project.

| Task | Owner | Due Date |

Task	Owner	Due Date
Data quality assessment and cleaning	Kirandeep Kaur	Feb 3
Feature engineering (customer metrics, dealer KPIs)	Manpreet & Kirandeep	Feb 6

Exploratory data analysis and visualization	All Team Members	Feb 6
Clustering implementation (dealers)	Ramon Rabusa	Feb 9
Customer segmentation clustering	Ramon Rabusa	Mar 7
Demo 1 presentation and materials	All Team Members	Feb 9
Statistical hypothesis testing (loyalty impact)	Ramon Rabusa	Feb 20
Business insights and ML-business linkage	Manpreet Kaur Loyal	Feb 24
Phase 1 Report documentation	All Team Members	Feb 26
Cluster optimization and validation	Ramon Rabusa	Mar 4
Demo 2 preparation and rehearsal	All Team Members	Mar 9
Interactive dashboard development	Kirandeep Kaur	Mar 28
Business recommendations formulation	Manpreet Kaur Loyal	Mar 17
Phase 2 Report documentation	All Team Members	Mar 19
Final presentation development	All Team Members	Apr 5
GitHub repository documentation	All Team Members	Ongoing

6. Resources Required:

List the resources necessary for completing the project.

Resource	Description	Estimated Cost
Google Colab	Cloud-based Python environment for ML implementation and collaboration	$0 (Free tier)
Python Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, plotly for data analysis and ML	$0 (Open source)
GitHub	Version control and code repository management	$0 (Free tier)
Slack Workspace	Team communication and coordination platform	$0 (Free tier)
Google Drive	Dataset storage and document collaboration	$0 (Free tier)
GO Auto Dataset	242,818 service records (2020-2024) provided by course client	$0 (Provided)
Team Members	Three CMPT 3830 students contributing approximately 10-15 hours per week each	$0 (Course work)
Personal Computers	Individual team member laptops for development and analysis	$0 (Owned)

Total Project Budget: No external costs.

7. Risk Management Plan:

Risk	Likelihood	Impact	Mitigation Strategy

Data quality issues or missing values affecting analysis	Medium	High	Conduct early data quality assessment (Week 1); develop robust cleaning protocols; document all data assumptions and limitations
Clustering models fail to produce meaningful segments	Medium	High	Test multiple clustering algorithms; iterate on feature selection; validate with domain knowledge and business logic
Team member unavailability due to illness or conflicts	Medium	Medium	Cross-train on key tasks; maintain clear documentation; require 48-hour advance notice for absences; redistribute work during weekly sync
Deadline conflicts with other courses	High	Medium	Begin major deliverables 1 week early; use buffer days in timeline; prioritize critical path tasks; communicate workload in weekly standups
Scope creep beyond course requirements	Medium	Medium	Reference project charter regularly; maintain 'in scope/out of scope' list; defer enhancement ideas to 'future work' section; prioritize deliverables over extras
Technical difficulties with tools or platforms	Low	Medium	Use established, reliable tools; maintain local backups; document environment setups; have

			fallback options (e.g., local Jupyter if Colab fails)
Insufficient business context for recommendations	Low	Medium	Prepare strategic questions for client meetings; research automotive service industry best practices; validate assumptions with instructor; focus on data-driven insights

Risk Review Schedule: Team will reassess risks during mid-project check-in (Week 7: February 24-27) and adjust mitigation strategies as needed.

9. Budget:

Total Project Budget: No external cost

This project utilizes entirely free and open-source resources. All required software (Python, machine learning libraries, Google Colab, GitHub, Slack, Google Drive) is available at no cost. The dataset is provided by the course client, and team members contribute their time as part of CMPT 3830 course requirements. No hardware purchases, software licenses, or external services requiring payment are necessary for successful project completion.

While the project has zero monetary cost, the team acknowledges significant time investment representing the primary 'cost' in terms of student effort and learning commitment.